

Comprehensive Coursework

Detailed notes compiled and verified with interactive models.

1. Intro & Material Science

Material Science represents a multidisciplinary study that bridges **Physics** and **Chemistry** to create a unified **Physical Science** framework. The primary goal of this discipline is to establish a rigorous, mathematical, and observational link between the microscopic structure of materials and their macroscopic properties.

Physics + Chemistry → Physical Science

By engineering the atomic-scale details of a solid, we directly influence its macroscopic behavior—such as its mechanical strength, electrical conductivity, refractive index, and thermal expansion. This is known as the **Structure-Property Correlation**.

Coursework Schedule Reference

The schedule details the designated periods dedicated to analyzing these properties:

Day	Period	Subject Focus
Monday	1	Solid State Theory
Tuesday	5	Materials Engineering
Wednesday	4	Solid State Theory
Thursday	1, 2	Solid State Theory (Extended Lab/Lecture)
Friday	4, 5	Materials Engineering (Applications)

2. Fundamental Physics of Solids

The physics of solids is divided into two primary axes of study: establishing the exact geometrical arrangement (**Structure**) and measuring or calculating the response to external forces/fields (**Property**).

$$\text{Physics of Solids} \longrightarrow \begin{cases} \text{Structure} \\ \text{Property} \end{cases}$$

Solids are characterized by a **strict geometry**. This geometry is a direct consequence of thermodynamic stability: *a physical system will always search for a state of minimum free energy to maximize stability*.

Fundamental Principles Involved

The study of solids is built upon three pillar principles of modern physics:

I. Energy Conservation

Lattice dynamics, phase changes, and electronic transitions are governed by thermodynamic conservation, where potential binding energy balances kinetic vibrational energy.

II. Momentum Conservation

Scattering processes (like X-ray or neutron diffraction) in a periodic lattice conserve momentum modulo a reciprocal lattice vector (crystal momentum).

III. Uncertainty Principle

Nature imposes a fundamental uncertainty limit on physical systems ($\Delta x \cdot \Delta p \geq \hbar/2$). On the atomic scale, matter and energy exist simultaneously: localized electrons represent mass, while delocalized waves/radiation represent energy.

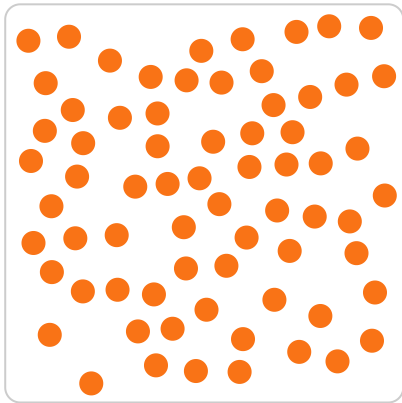
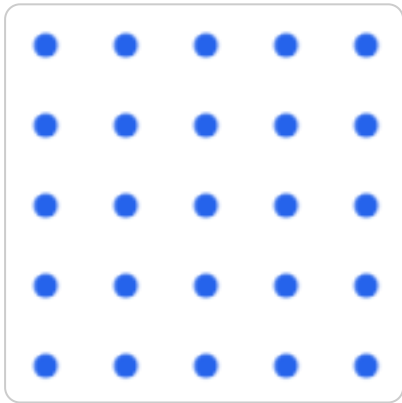
Note on Wave Mechanics: The primary function of a wave in a solid is to propagate and carry energy through **spacetime** without any net transport of the physical matter itself.

3. Crystalline vs. Amorphous Solids

Solids are classified based on the spatial order of their constituent particles. Crystalline materials possess long-range periodic order, while amorphous materials lack this long-range periodicity.

Property	Crystalline Solids	Amorphous Solids
Ordering	Long-range periodic order (extends over thousands of atomic distances).	Short-range order only (extends only to ~5 atomic spacings, denoted by $a_0 \rightarrow \approx 5$).
Directionality	Anisotropic: Physical properties (e.g., refractive index, conductivity) vary with measuring direction.	Isotropic: Physical properties are identical in all directions due to spatial randomness.

Property	Crystalline Solids	Amorphous Solids
Thermodynamic State	Low energy, thermodynamically stable state.	High energy, metastable state.
Cleavage & Fracture	Cleaves along flat, distinct crystallographic planes.	Fractures along random, curved paths (conchoidal fracture). *e.g., glass breaks differently every time.*
Randomness	Minimum randomness (entropy).	High randomness (entropy).

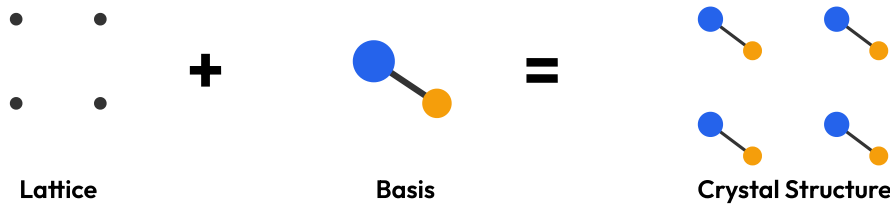


Visual comparison of Crystalline (left, regular grid) vs. Amorphous (right, disordered with short-range order) configurations.

4. Crystal Lattice & Basis

A physical crystal structure is mathematically constructed by associating a physical group of atoms (the **Basis**) with an infinite geometric array of points (the **Lattice**).

$$\text{Lattice} + \text{Basis} = \text{Crystal Structure}$$



Primitive Unit Cell: A unit cell that contains exactly 1 effective lattice point ($n_{\text{eff}} = 1$). For example, a simple cubic (SC) cell has 8 corner points, each shared by 8 adjacent cells:

$$n_{\text{eff}} = 8 \times \frac{1}{8} = 1$$

Non-Primitive Unit Cell: A unit cell containing more than 1 effective lattice point ($n_{\text{eff}} \neq 1$). In Body-Centered Cubic (BCC) crystals, $n_{\text{eff}} = 2$, and in Face-Centered Cubic (FCC) crystals, $n_{\text{eff}} = 4$.

5. Electron Wavelength & Zero-Point Energy

To probe the atomic lattices of solids, we employ diffraction techniques using waves whose wavelengths match the characteristic interatomic spacing ($d \approx 1 - 2 \text{ \AA}$).

$$\lambda = \frac{h}{p}, \quad E = h\nu$$

Wavelength Calculation for $E = 50 \text{ eV}$

If we model a photon of energy $E = 50 \text{ eV}$, its wavelength is calculated as:

$$\lambda = \frac{hc}{E} = \frac{6.626 \times 10^{-34} \text{ J} \cdot \text{s} \times 3 \times 10^8 \text{ m/s}}{50 \times 1.602 \times 10^{-19} \text{ J}} \approx 24.75 \text{ nm}$$

In the handwritten notes, a rapid approximation $6.6 \approx 6.4$ was used to simplify divisions:

$$\lambda \approx \frac{6.4 \times 3}{1.6 \times 50} \times 10^{-7} \text{ m} = \frac{4 \times 3}{50} \times 10^{-7} \text{ m} = \frac{6}{25} \times 10^{-7} \text{ m} = 24 \text{ nm}$$

Quantum Harmonic Oscillator & Zero-Point Motion

According to quantum mechanics, the atoms in a solid lattice behave like harmonic oscillators. The minimum possible energy of a quantum harmonic oscillator is not zero (unlike classical mechanics), but is bounded by the uncertainty principle:

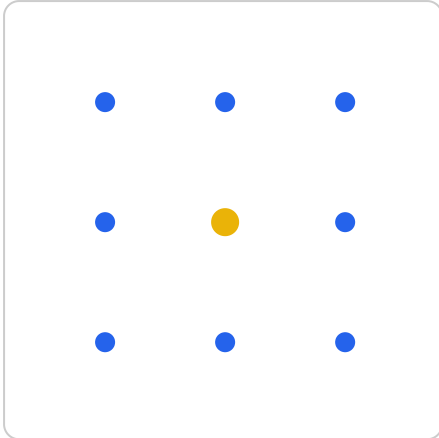
$$E_0 = \frac{1}{2} h\nu$$

Because this ground-state energy (zero-point energy) is strictly positive ($E_0 > 0$), **the atoms in a crystal are never at absolute rest at any position, even at absolute zero (0 K).**

6. Wigner-Seitz Unit Cell

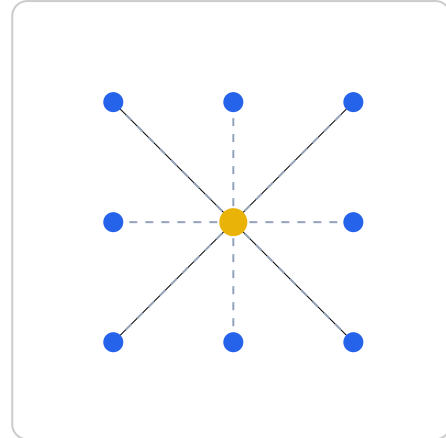
A Wigner-Seitz cell is a geometric method to define a primitive unit cell which preserves the full point symmetry of the underlying Bravais lattice.

Step 1



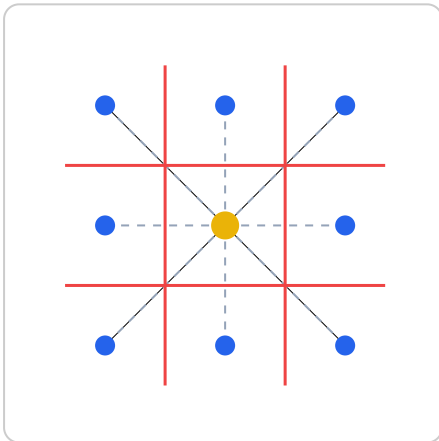
Identify a central lattice point (gold).

Step 2



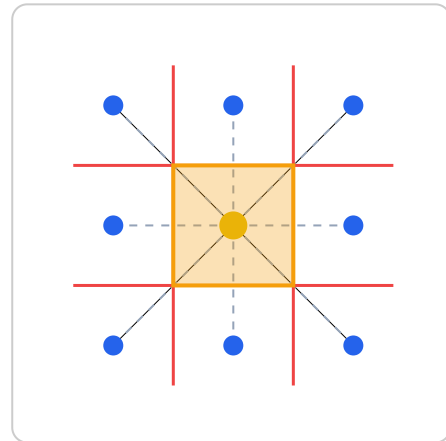
Connect the central point to all nearest neighbors with vector offsets.

Step 3



Construct perpendicular bisectors (normals) at the midpoints of the lines.

Step 4



The smallest enclosed volume forms the primitive Wigner-Seitz cell.

7. Symmetry Operations & Group Theory

A symmetry operation maps the crystal lattice onto itself. These operations are categorized into:

Translation: Pure vector offset $\vec{T} = n_1\vec{a}_1 + n_2\vec{a}_2 + n_3\vec{a}_3$ where $n_i \in \mathbb{Z}$ for primitive cells.

Rotation: Rotates the crystal around an axis. Only specific angles ($360^\circ/n$) are compatible with translation periodicity.

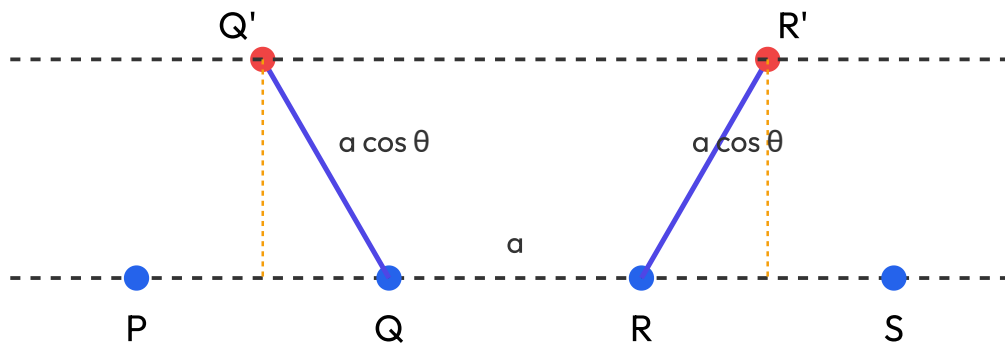
Reflection: Mirrors coordinates across a mirror plane (m).

Inversion: Points are inverted through a central point ($\vec{r} \rightarrow -\vec{r}$). Only possible in 3D.

Symmetry operations that leave at least one point invariant are grouped into **Point Groups**. There are exactly **10 point groups in 2D** and **32 point groups in 3D**.

8. Proof: 5-Fold Symmetry Contradiction

Crystals cannot have 5-fold (or any ≥ 7 -fold) rotational symmetry. The following proof derives this limitation mathematically by demanding compatibility with translation symmetry.



Geometry of the rotational symmetry proof. Points Q' and R' must be lattice points.

Consider a row of collinear lattice points spaced by distance a . Rotate point R about Q counter-clockwise by θ , and rotate Q about R clockwise by θ . The resulting points Q' and R' must also be lattice points. The distance $Q'R'$ must be an integer multiple of a :

$$Q'R' = a + 2a \cos \theta = m \cdot a \quad (m \in \mathbb{Z})$$

This simplifies to:

$$\cos \theta = \frac{m - 1}{2}$$

Since $-1 \leq \cos \theta \leq 1$, m can only take integer values between -1 and 3 :

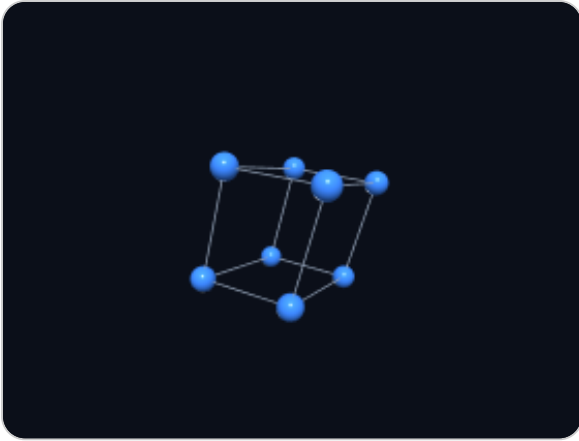
m	$\cos \theta$	Allowed Angle θ	Symmetry Fold $n = 2\pi/\theta$
-1	-1	180°	2 (Diad)
0	$-1/2$	120°	3 (Triad)
1	0	90°	4 (Tetrad)
2	$1/2$	60°	6 (Hexad)
3	1	360°	1 (Monad)

Since no integer m gives $\cos \theta = \cos(72^\circ) \approx 0.309$, **5-fold rotational symmetry is not allowed in periodic crystals.**

9. Bravais Lattices & Crystal Systems

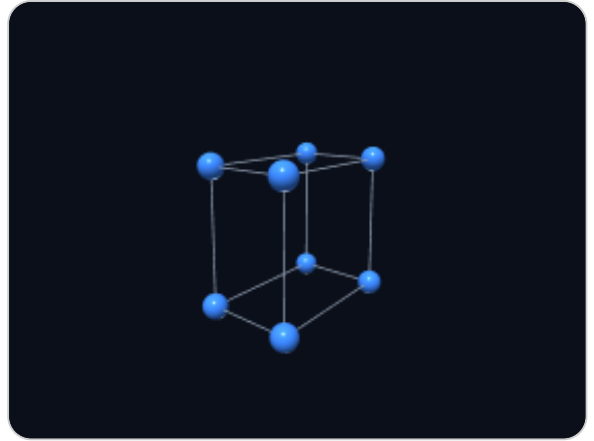
Crystal structures are classified into **7 Crystal Systems** based on their axial relationships and interaxial angles. In 3D space, these yield exactly **14 Bravais Lattices**.

1. Triclinic (Simple / P)



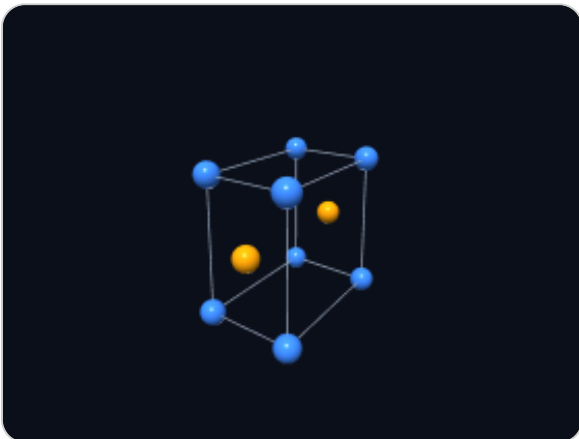
$$a \neq b \neq c$$
$$\alpha \neq \beta \neq \gamma \neq 90^\circ$$

2. Monoclinic (Simple / P)



$$a \neq b \neq c$$
$$\alpha = \gamma = 90^\circ, \beta \neq 90^\circ$$

3. Monoclinic (Base-Centered / C)



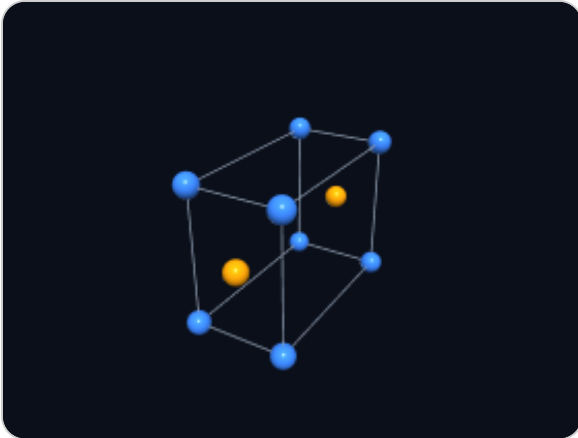
$$a \neq b \neq c$$
$$\alpha = \gamma = 90^\circ, \beta \neq 90^\circ$$

4. Orthorhombic (Simple / P)



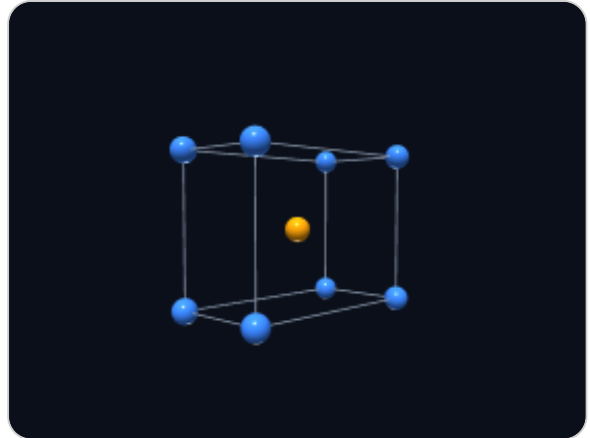
$$a \neq b \neq c$$
$$\alpha = \beta = \gamma = 90^\circ$$

5. Orthorhombic (Base-Centered / C)



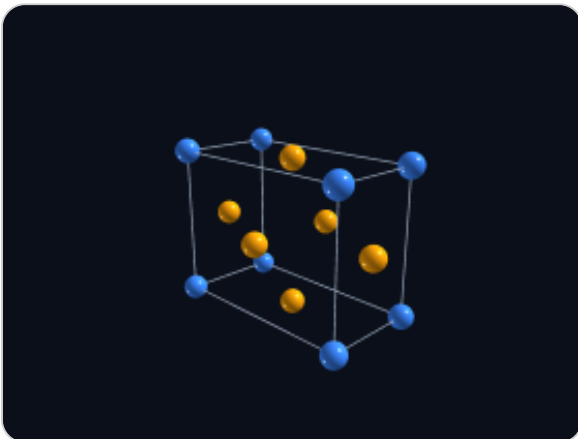
$$a \neq b \neq c$$
$$\alpha = \beta = \gamma = 90^\circ$$

6. Orthorhombic (Body-Centered / I)



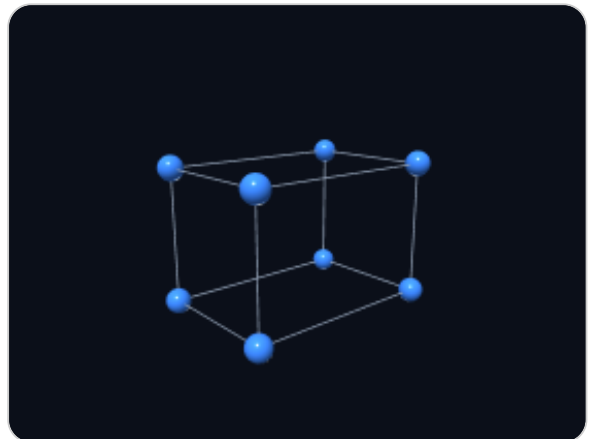
$$a \neq b \neq c$$
$$\alpha = \beta = \gamma = 90^\circ$$

7. Orthorhombic (Face-Centered / F)



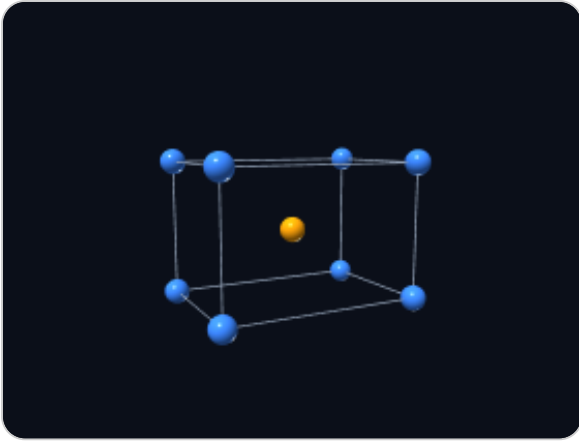
$$a \neq b \neq c$$
$$\alpha = \beta = \gamma = 90^\circ$$

8. Tetragonal (Simple / P)



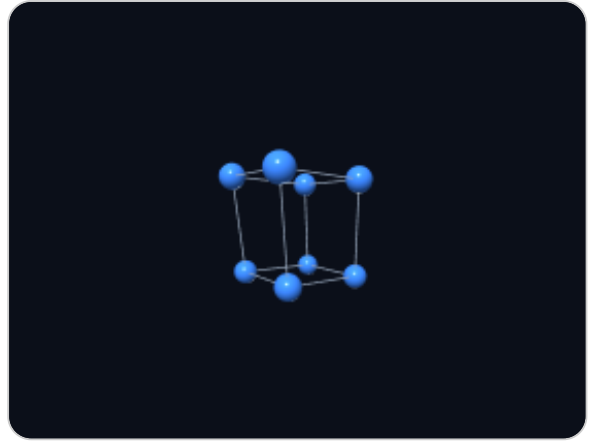
$$a = b \neq c$$
$$\alpha = \beta = \gamma = 90^\circ$$

9. Tetragonal (Body-Centered / I)



$$a = b \neq c$$
$$\alpha = \beta = \gamma = 90^\circ$$

10. Rhombohedral (R / Trigonal)



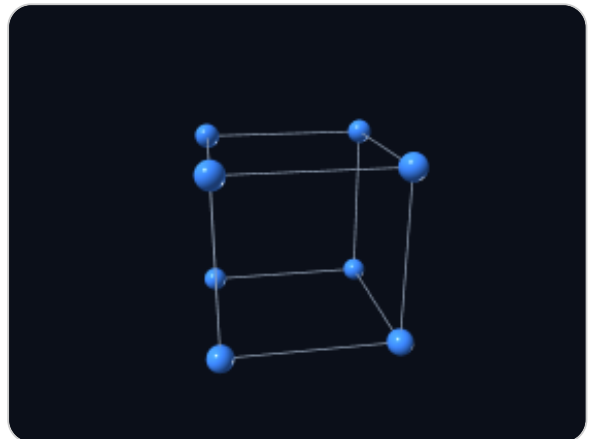
$$a = b = c$$
$$\alpha = \beta = \gamma \neq 90^\circ$$

11. Hexagonal (Simple / P)



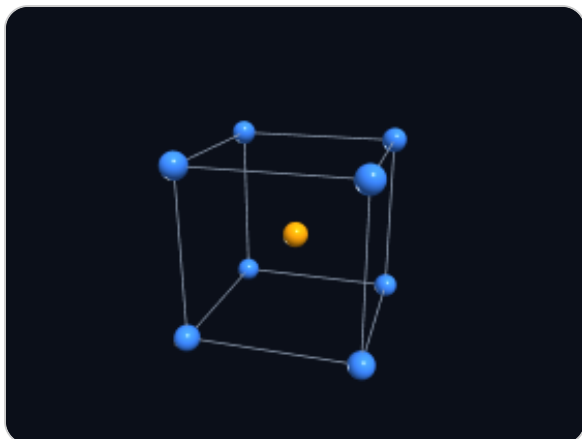
$$a = b \neq c$$
$$\alpha = \beta = 90^\circ, \gamma = 120^\circ$$

12. Simple Cubic (SC / P)



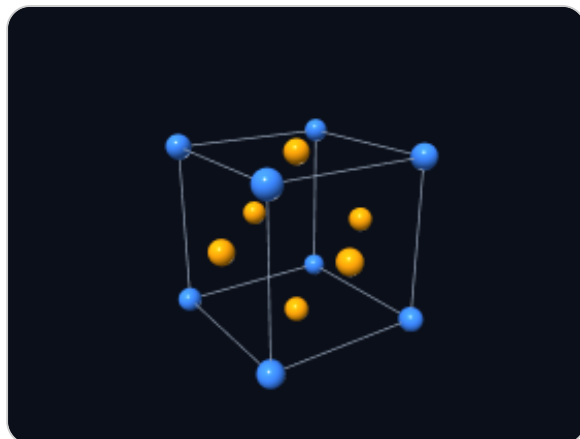
$$a = b = c$$
$$\alpha = \beta = \gamma = 90^\circ$$

13. Body-Centered Cubic (BCC / I)



$$a = b = c$$
$$\alpha = \beta = \gamma = 90^\circ$$

14. Face-Centered Cubic (FCC / F)



$$a = b = c$$
$$\alpha = \beta = \gamma = 90^\circ$$

The 7 3D Crystal Systems

Crystal System	Axial Lengths	Interaxial Angles	Bravais Lattices
Cubic	$a = b = c$	$\alpha = \beta = \gamma = 90^\circ$	Simple (P), Body-Centered (I), Face-Centered (F)
Tetragonal	$a = b \neq c$	$\alpha = \beta = \gamma = 90^\circ$	Simple (P), Body-Centered (I)
Orthorhombic	$a \neq b \neq c$	$\alpha = \beta = \gamma = 90^\circ$	Simple (P), Base-Centered (C), Body-Centered (I), Face-Centered (F)
Monoclinic	$a \neq b \neq c$	$\alpha = \gamma = 90^\circ, \beta \neq 90^\circ$	Simple (P), Base-Centered (C)
Triclinic	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90^\circ$	Simple (P)
Hexagonal	$a = b \neq c$	$\alpha = \beta = 90^\circ, \gamma = 120^\circ$	Simple (P)
Rhombohedral	$a = b = c$	$\alpha = \beta = \gamma \neq 90^\circ$	Simple (P)

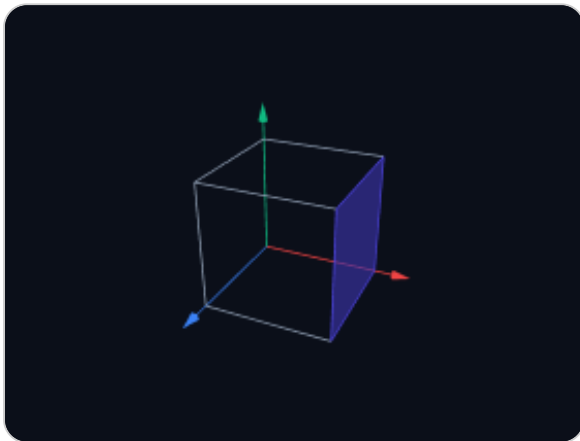
10. Miller Indices of Planes & Directions

Crystallographic directions and planes are represented by coordinates enclosed in specific brackets. A direction vector is denoted as $[hkl]$, and a plane is denoted as (hkl) .

Recipe to Determine Miller Indices (hkl) of a Plane

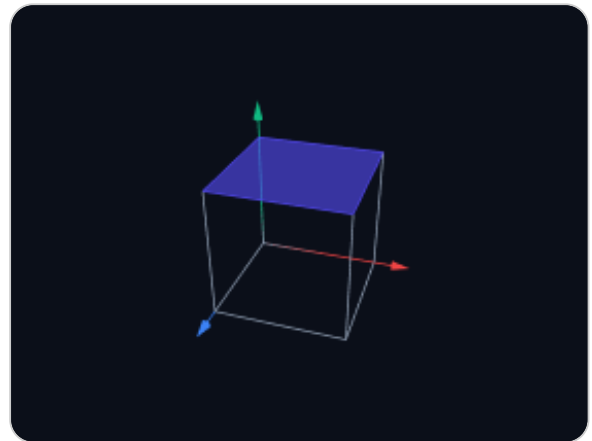
1. Find the intercepts of the plane with the coordinate axes (x, y, z) in terms of lattice constants.
2. Take the reciprocals of these intercepts.
3. Clear fractions by multiplying by the lowest common multiple (LCM) to get integers.
4. Enclose the integers in parentheses: (hkl) . Write a bar over negative integers (e.g. $\bar{1}$).

Plane (100)



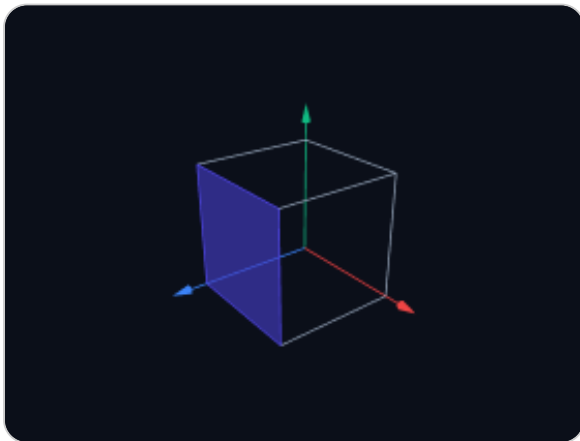
$$x = 1$$

Plane (010)



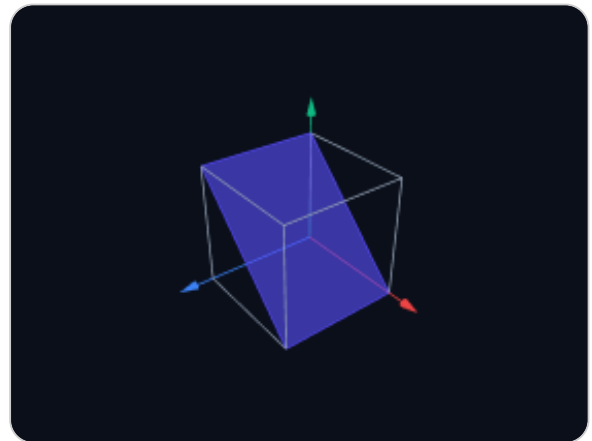
$$y = 1$$

Plane (001)



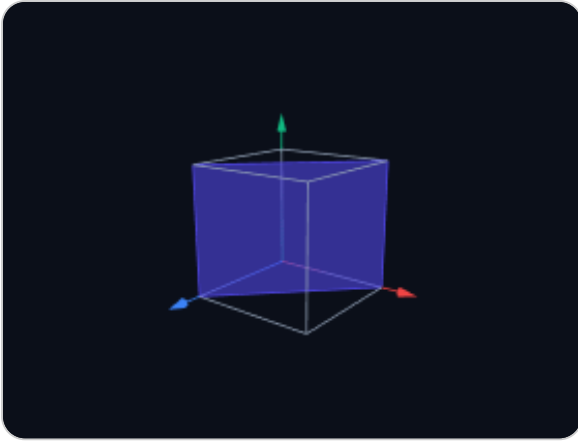
$$z = 1$$

Plane (110)



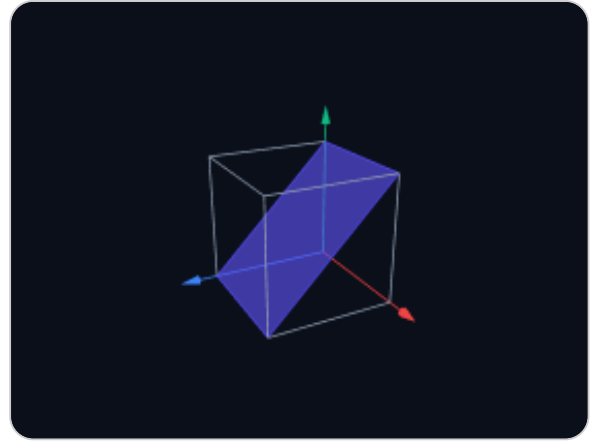
$$x + y = 1$$

Plane (101)



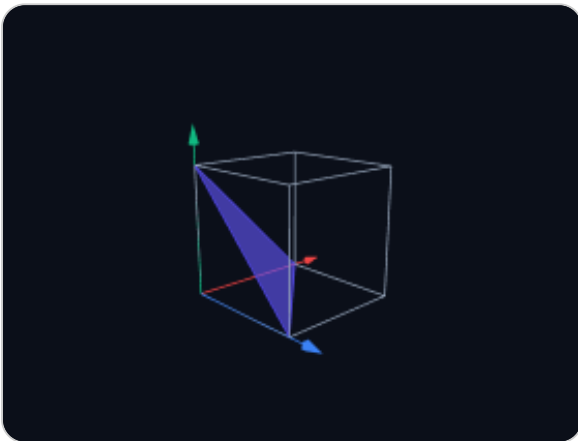
$$x + z = 1$$

Plane (011)



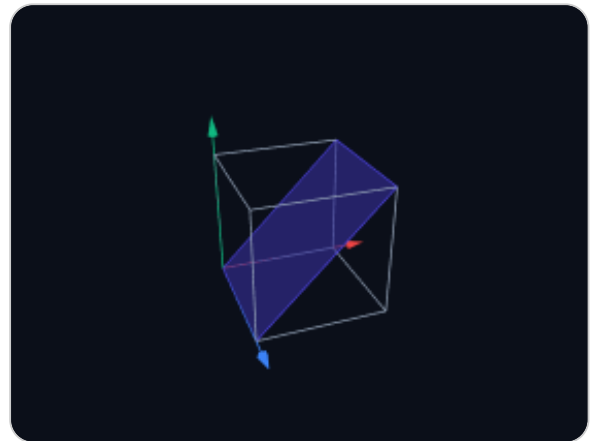
$$y + z = 1$$

Plane (111)



$$x + y + z = 1$$

Plane ($\bar{1}10$)



$$x - y = 1$$

11. Interplanar Spacing (d_{hkl})

The interplanar spacing d_{hkl} is the perpendicular distance between adjacent parallel planes with Miller indices (hkl) .

Derivation for Orthogonal Lattices

Consider a plane that intersects the axes of an orthogonal coordinate system at $x = a/h$, $y = b/k$, and $z = c/l$. The normal vector from the origin to this plane has length d and

makes angles $\theta_1, \theta_2, \theta_3$ with the axes:

$$\cos \theta_1 = \frac{d}{a/h} = \frac{hd}{a}, \quad \cos \theta_2 = \frac{kd}{b}, \quad \cos \theta_3 = \frac{ld}{c}$$

Using the geometric constraint for direction cosines ($\cos^2 \theta_1 + \cos^2 \theta_2 + \cos^2 \theta_3 = 1$):

$$d^2 \left(\frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2} \right) = 1 \implies d_{hkl} = \frac{1}{\sqrt{\frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}}}$$

Spacing Formulas by System

Cubic ($a = b = c$)

$$d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

Tetragonal ($a = b \neq c$)

$$d = \frac{ac}{\sqrt{c^2(h^2 + k^2) + a^2l^2}}$$

Orthorhombic ($a \neq b \neq c$)

$$d = \frac{1}{\sqrt{\frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}}}$$